

DECODABLE IS NOT DIRECTION-SPECIFIC: CAUSAL TESTS AT THE FINAL VISUAL BLOCK OF MEDICAL VISION-LANGUAGE MODELS

Anonymous authors

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ABSTRACT

A linear probe can reveal a clinical label without showing that its decoded direction selectively affects a model’s answer. We test this distinction at the exact final visual block consumed downstream by LLaVA-1.5-7B and Qwen2.5-VL-7B. Our registered protocol combines a fixed 512-dimensional readout, patient-disjoint evaluation, type-to-random-label controls, patient-bootstrap uncertainty, and relative-token probe-normal interventions against random, sham, and fixed clinical directions. Across three model–concept cells on NIH ChestX-ray14, AUROC ranges from 0.7742 to 0.8009 and every selectivity interval excludes zero. Neither LLaVA Effusion nor LLaVA Edema clears random/sham controls. Qwen Effusion clears them twice, changing mean $P(\text{yes})$ by 0.1945 on 400 independent patients, but Nodule changes the same answer by 0.2519; the familywise Effusion-minus-maximum-unrelated margin is -0.0573 (95% CI $[-0.0694, -0.0450]$). Two prospective mechanism gates sharpen this boundary. A six-direction by six-question Qwen matrix yields zero owned concepts and no shared alias beyond 119 random directions. A fresh 50-patient Consolidation test finds positive controlled selectivity but no reliable input-displacement closure: the concept gain is 0.0022 versus a random maximum of 0.0025. Thus, in these registered cells, linear availability at a consumed visual locus does not establish concept-specific causal use under probe-normal intervention. The experiments separate decodability, generic perturbation sensitivity, clinical direction specificity, and mechanism confirmation.

1 INTRODUCTION

A probe can name a clinical signal in a representation. The consequential question is whether changing that named direction selectively changes the model’s answer. Linear probes are useful because they expose information available to a restricted readout family (Alain & Bengio, 2017; Hewitt & Liang, 2019). Their accuracy alone, however, does not establish that the underlying model uses the decoded property for its task (Ravichander et al., 2021).

This distinction matters acutely in medical vision-language models (VLMs). A high clinical probe score is easy to read as evidence that a finding is operational inside the model, even though the score may reflect probe capacity, acquisition structure, or information that the downstream computation ignores. Recent medical VLM work traces classification signals across depth (Zhu et al., 2026), while VLM steering work uses probe vectors to edit visual concepts (Theodoridis et al., 2026). These lines establish that clinical or visual properties can be read and that internal edits can move outputs. They leave open whether a target probe normal has a *direction-specific* effect when compared with equally strong, clinically meaningful alternatives at the same consumed locus.

We ask a narrow causal question: when a clinical concept is robustly linearly decodable at the final visual block, does its probe normal exert direction-specific influence on the answer? Figure 1 summarizes the protocol. We first verify the exact final visual block on the model’s forward path. At that locus, we fit one capacity-matched probe under a patient split and compare its AUROC with 20 type-to-random-label controls. We then add the unit probe normal to each visual token at a dose proportional to that token’s activation norm. Random directions, a coordinate-permutation sham, and

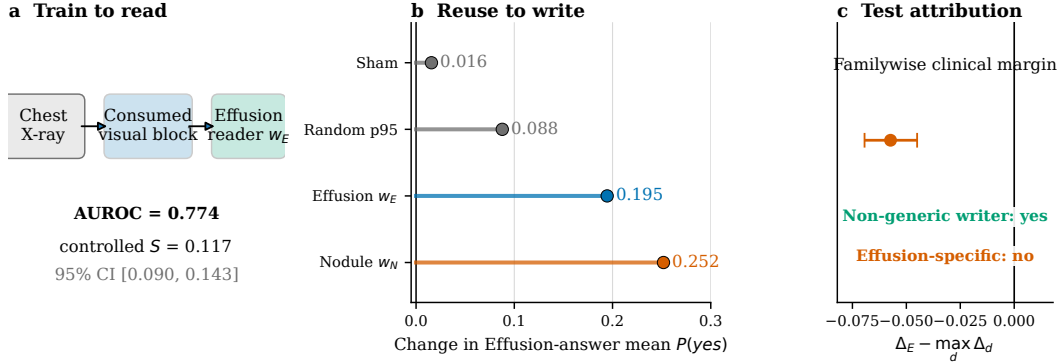


Figure 1: **Decodability, generic perturbation selectivity, and clinical direction specificity are distinct tests.** (a) A fixed-capacity probe reads a clinical label from the exact final visual block consumed downstream; equal-norm interventions edit that same representation. (b) All three registered cells have positive controlled-decoding intervals under patient bootstrap. (c) Filled circles mark passed rungs; crosses mark where a cell stops. LLaVA Effusion and Edema stop at random/sham. Qwen Effusion clears that rung, then stops at fixed clinical directions on an independent 400-patient cohort. Open circles mark later rungs not tested for the two LLaVA cells.

fixed unrelated clinical probe normals receive the same dose. The design treats availability, generic perturbation selectivity, and clinical direction specificity as successive evidentiary rungs.

Our evidence begins with three registered model–concept cells on NIH ChestX-ray14. All three target labels are linearly available, with AUROC between 0.7742 and 0.8009 and positive controlled selectivity. The two LLaVA cells do not clear random/sham controls. Qwen Effusion clears those controls on a prospectively selected cohort of 400 new patients, but the fixed Nodule direction produces a larger response and the familywise Effusion-minus-maximum-unrelated interval lies entirely below zero. We then test mechanism more directly. A six-direction by six-question Qwen matrix yields no owned clinical concept or shared alias beyond 119 random directions, and a fresh within-patient Consolidation test does not close the measured input gap beyond its registered controls.

The paper makes three contributions:

- We introduce a forward-path, capacity-controlled protocol that separates controlled decodability, random/sham intervention selectivity, and clinical direction specificity at one consumed representation.
- We report three final-block cells in which controlled decodability is positive while the registered direction-specific endpoint is not reached.
- We replicate Qwen’s behavioural response on independent patients, then find zero owned concepts in a prospective 6×6 direction–question matrix and no controlled input closure for Consolidation.

The resulting principle is practical: a decoded clinical direction warrants a concept-specific causal interpretation only after matched controls establish specificity and the proposed read–write mechanism survives direct tests at the same locus and dose. Sections 2–3 place and define these tests, Section 4 reports the evidence chain, and Sections 5–6 give its interpretation and scope.

2 RELATED WORK

Controlled probing. Linear classifiers have long been used to track information across network depth (Alain & Bengio, 2017). A probe score reflects both the representation and the learner fitted on top of it, motivating matched-capacity probes and control tasks (Hewitt & Liang, 2019). The broader probing literature distinguishes information that a readout can recover from information that a model uses for its task (Belinkov, 2022). On synthetic tasks, high probing accuracy can persist for properties that the model provably does not need, directly separating decodability from task relevance

(Ravichander et al., 2021). Causal erasure methods pursue the complementary test: remove linearly identified information and measure the behavioural consequence (Ravfogel et al., 2020; Elazar et al., 2021). Our protocol brings these concerns to medical images with patient-disjoint splits, a common 512-dimensional projection, repeated recurring-type controls, and bootstrap intervals. It asks about one verified consumed visual locus rather than selecting the best layer after observing test results.

Representation interventions. Activation engineering changes intermediate representations at inference time and measures how outputs respond (Turner et al., 2023; Panickssery et al., 2024). A probe normal is an appealing intervention direction because it supplies a semantic label and a signed vector from the same representation. Yet a behavioural change under that vector can follow from generic sensitivity to perturbation magnitude, a coordinate-dependent construction, or overlap with another concept direction. Random vectors and a coordinate-permutation sham address the first two possibilities. Fixed clinical probe normals address the third by asking whether the target direction is more effective than plausible semantic alternatives. Our primary statistic recomputes the maximum unrelated clinical effect inside each patient-bootstrap replicate, so the direction-specific comparison is one familywise endpoint rather than five separate pairwise claims.

Intervention families answer different causal questions. Iterative nullspace projection and amnesic probing test whether removing linearly recoverable information damages a downstream task (Ravfogel et al., 2020; Elazar et al., 2021). Additive steering instead tests whether a chosen direction can write a change into the forward pass (Turner et al., 2023). The second test needs an operating scale: an unnormalized vector can appear effective simply because its norm is large, while a fixed Euclidean step can represent very different perturbations across layers or architectures. We use each token’s own activation norm as the scale and compare every direction at the same α . This choice makes the random, sham, target, and clinical interventions commensurate within a registered cell. It also keeps a generic sensitivity result separate from a claim that the probe normal uniquely represents its named clinical concept.

Medical and multimodal VLM interpretation. Depth-wise probing of medical multimodal models has identified losses and gains of classification information across vision towers, connectors, language layers, and semantic outputs (Zhu et al., 2026). Layer-wise probes of MedCLIP also expose acquisition and device shortcuts in chest radiographs (Pedersen et al., 2026). These studies motivate tight nuisance and data controls, but they do not intervene on the decoded clinical direction at the same locus. Conversely, controlled visual-concept experiments in lightweight driving VLMs combine probing with probe-vector steering and identify cases where information is linearly present but generation still fails (Theodoridis et al., 2026). Targeted edits in general VLMs likewise connect linearly available visual features to downstream predictions (Rajaram et al., 2025). We focus on 7B medical-question answering, preserve the clinical patient split, and compare the target probe normal with equal-norm random, sham, and clinical directions.

Together, this literature motivates three separate questions. Controlled probing asks whether a restricted readout reliably recovers a label. Generic intervention controls ask whether the probe normal moves behaviour beyond non-semantic perturbations. Clinical-direction controls ask whether the response is specific to the named finding. Our experiments align these questions at the same forward-path representation.

3 REGISTERED MEASUREMENT PROTOCOL

Data and split. We use NIH ChestX-ray14 (Wang et al., 2017), whose findings are mined from radiology reports. A deterministic hash of patient identity assigns rows to disjoint train, validation, and test partitions. The registered manifest contains 18,212 training images and 5,343 test images from 1,659 test patients. The three cells pair LLaVA-1.5-7B with Effusion and Edema, and Qwen2.5-VL-7B with Effusion. All reported uncertainty clusters by patient. Prompts and exact run identifiers appear in Appendix A.

Models and consumed loci. For LLaVA-1.5-7B (Liu et al., 2024), `vis.last` is the output of `model.vision_tower.encoder.layers.22`; for Qwen2.5-VL-7B (Bai et al., 2025), it is `model.visual.blocks.31`. Synthetic hook tests establish that each module is on the downstream path, that an $\alpha = 0$ intervention is bitwise identical to the unmodified forward pass, and that a nonzero

intervention changes the connector or merger and final logits. The probe and intervention therefore operate on the same final visual representation that downstream computation consumes.

Controlled decodability. For image i , mean pooling over the final-block visual tokens yields $\mathbf{h}_i \in \mathbb{R}^D$. A seed-0 Gaussian projection maps every model to 512 dimensions; the scaler is fitted on training rows only. We then fit logistic regression with $C = 1$, maximum 2,000 iterations, and seed 0. The real-label AUROC is compared with 20 control AUROCs. Each control seed maps the same recurring input type—AP view indicator, sex indicator, and age decade, giving 39 strata—to a random label. Selectivity is

$$S = \text{AUROC}(y, \hat{y}) - \frac{1}{20} \sum_{s=0}^{19} \text{AUROC}(\tilde{y}_s, \hat{y}_s). \quad (1)$$

We report the real AUROC, control distribution, and S with 2,000 patient-cluster bootstrap resamples. Paired contrasts reuse identical ordered test rows and bootstrap draws.

Probe-normal intervention. Let $\hat{\mathbf{w}}_c$ be the unit normal of the capacity-matched probe for concept c . At every visual token t , the registered intervention is

$$\mathbf{h}'_{it} = \mathbf{h}_{it} + \alpha \|\mathbf{h}_{it}\|_2 \hat{\mathbf{w}}_c. \quad (2)$$

This relative-token scale holds perturbation magnitude fixed as a fraction of the local activation norm. The endpoint is the paired change in mean $P(yes)$ from the common unsteered baseline. Original cells use concept doses $\{-1, -0.5, -0.25, -0.1, 0, 0.1, 0.25, 0.5, 1\}$ on 200 held-out test images. Controls are evaluated at the six nonzero doses excluding ± 0.1 : 20 isotropic random unit directions, one coordinate-permutation sham, and fixed unrelated clinical probe normals. Dose monotonicity over $[-0.5, 0.5]$ is descriptive.

Original-cell decision rule. For each controlled nonzero dose, we orient the raw change by the sign registered for the concept direction. The primary effect is the maximum concept-consistent change over those doses. A cell clears the generic intervention rung only if this effect leaves the same-dose random 5th–95th percentile interval in the concept-consistent direction and exceeds the maximum absolute sham effect. The rule is fixed across the three cells.

Direction-specificity supplement. The original Qwen cell motivated a prospective independent-row test at its discovery-locked dose, $\alpha = +0.25$. We exclude all patients in the original 200-image cohort, select one row per remaining test patient by a label-blind hash, rank patients by a second label-blind hash, and take 400. The grid contains a common unsteered baseline, Effusion, 20 random directions, the sham, and five fixed clinical directions: Atelectasis, Pneumothorax, Cardiomegaly, Mass, and Nodule. It produces 28 aggregate configurations and 11,200 per-image outcomes.

For direction d , let Δ_d be its paired mean change from baseline. The familywise statistic is

$$M = \Delta_{\text{Eff}} - \max_{d \in \mathcal{D}_{\text{clinical}}} \Delta_d. \quad (3)$$

The maximum is recomputed inside each of 5,000 patient-bootstrap draws. Direction specificity requires a positive Effusion effect above random p95 and absolute sham, together with a one-sided 95% lower bound for M above zero.

Causal-ownership matrix. To test whether another clinical normal better matches the answer pathway, we prospectively cross six Qwen directions—Effusion, Atelectasis, Pneumothorax, Cardiomegaly, Mass, and Nodule—with their six matching questions on 400 new patients. Every column uses the same $\alpha = +0.25$, 119 random directions, and a sham. For question c , ownership requires the diagonal effect to exceed every other clinical direction, every random direction, and the absolute sham effect. A centered studentized max- T bootstrap gives simultaneous 95% lower bounds over all 30 clinical contrasts. The shared-alias family contains two statistics per direction—mean off-diagonal excess over the six named diagonals and mean absolute off-diagonal effect—for 12 max- T bounds; its effect must also exceed the global random and sham maxima.

Table 1: Controlled decodability at the exact final visual block: `CLIP encoder.layers.22` for LLaVA and `model.visual.blocks.31` for Qwen. Intervals are 95% patient-cluster bootstrap intervals over 2,000 draws. The control column gives the mean AUROC over 20 recurring-type random-label maps and their 5th–95th percentile spread.

Model	Concept	Locus	AUROC [95% CI]	Control mean (5–95%)	Selectivity [95% CI]
LLaVA-1.5-7B	Effusion	CLIP L22	0.7788 [0.7538, 0.8031]	0.6649 (0.5750–0.7450)	0.1139 [0.0875, 0.1404]
LLaVA-1.5-7B	Edema	CLIP L22	0.8009 [0.7648, 0.8329]	0.6649 (0.5750–0.7450)	0.1360 [0.0994, 0.1693]
Qwen2.5-VL-7B	Effusion	Qwen B31	0.7742 [0.7488, 0.7993]	0.6576 (0.5738–0.7476)	0.1166 [0.0897, 0.1428]

Within-patient input closure. When the matrix confirms neither ownership nor an alias, the registered fallback tests Consolidation on 50 fresh patients who each have one hash-selected positive and negative study. Let γ_i be the positive-minus-negative pooled `vis.last` displacement along the train-fitted Consolidation normal. The write dose is $\alpha_i = \text{clip}(\gamma_i / \|h_{it}\|, -0.25, 0.25)$ on the negative study. The closure gain is the reduction in its absolute $P(\text{yes})$ gap from the paired positive study. Confirmation requires a positive displacement lower bound, a concept gain above all 119 random directions and the sham, and a positive familywise lower bound against Effusion, Atelectasis, Pneumothorax, Cardiomegaly, Mass, and Nodule over 5,000 paired bootstraps.

Evidence and review. Every run uses an immutable commit snapshot and records commands, environment, timings, outputs, and terminal checksums. Internal validation replays decision summaries from accepted artifacts. A pinned, read-only cross-family reviewer checks each hard gate after internal verification; review can reject a gate but cannot create experimental evidence.

4 RESULTS: AVAILABILITY, SELECTIVITY, AND SPECIFICITY

4.1 CONTROLLED DECODABILITY IS ROBUST IN EVERY CELL

All three final-block representations make the target label available to the fixed linear readout (Table 1). LLaVA Effusion reaches AUROC 0.7788 and selectivity 0.1139 (95% CI [0.0875, 0.1404]). LLaVA Edema reaches AUROC 0.8009 and selectivity 0.1360 ([0.0994, 0.1693]). Qwen Effusion reaches AUROC 0.7742 and selectivity 0.1166 ([0.0897, 0.1428]). Each selectivity interval excludes zero despite the control maps’ substantial spread.

The paired decoding contrasts do not resolve a concept or architecture ordering. Within LLaVA, Edema-minus-Effusion selectivity is 0.0221 with a paired 95% CI of [−0.0175, 0.0590]. For Effusion, Qwen-minus-LLaVA selectivity is 0.0026 with a paired CI of [−0.0155, 0.0215]. The cells therefore provide similarly strong availability evidence without supporting a rank claim.

4.2 SIMILAR AVAILABILITY PRODUCES DIFFERENT RANDOM/SHAM OUTCOMES

Figure 2 shows the complete target dose curves and matched controls. LLaVA Effusion attains a maximum concept-consistent change of 0.1470 at $\alpha = -1$. This value exceeds the same-dose random p95 of 0.1202 but remains below the maximum absolute sham effect of 0.1546. The response is monotone over [−0.5, 0.5] (Spearman $\rho = 1.0$), yet monotonicity does not replace the failed sham criterion.

LLaVA Edema attains 0.0659 at $\alpha = -1$. It lies inside the same-dose random interval [−0.1304, 0.1470] and below the maximum absolute sham effect of 0.0913. Its descriptive dose monotonicity is $\rho = -0.4286$. Edema’s point estimate for controlled decoding is higher but statistically indistinguishable from Effusion’s, and it does not clear the generic intervention rule.

Qwen Effusion clears the original random/sham rung. Its maximum concept-consistent change is 0.2343 at $\alpha = +0.25$, above random p95 0.0963 and the maximum absolute sham effect 0.0852. The response is nearly monotone over the registered central range ($\rho = 0.9643$). This original rule

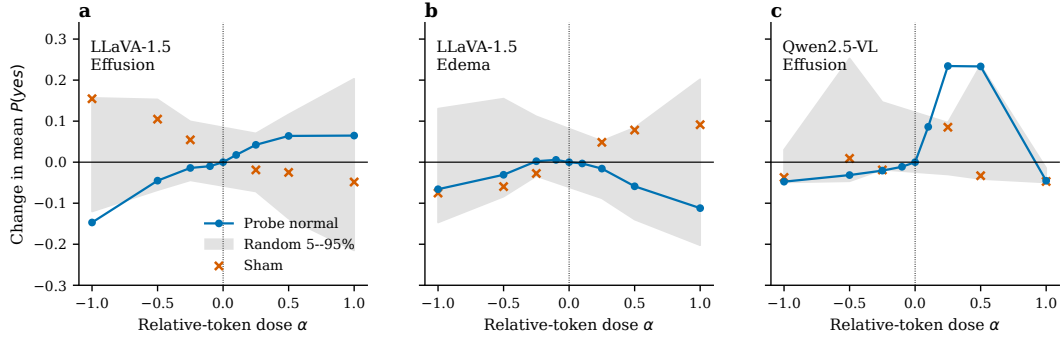


Figure 2: **Registered dose responses at the consumed final visual block.** The blue curve is the raw change in mean $P(\text{yes})$ under the target probe normal on the same 200 held-out images per cell. Gray bands show the 5th–95th percentile across 20 equal-norm random directions at the six controlled nonzero doses; orange crosses show the coordinate-permutation sham. The concept-consistent orientation used by the registered decision rule accounts for the negative-dose maxima in the two LLaVA cells.

Table 2: Registered intervention outcomes. Original cohorts select the maximum concept-consistent effect over the six controlled nonzero doses; their clinical-direction endpoint was not prospectively familywise. The independent cohort locks $\alpha = +0.25$ and recomputes the maximum over five fixed clinical directions inside each of 5,000 patient-bootstrap replicates. “Stopped at” names the first evidentiary rung not cleared.

Generic-control outcomes							
Model	Concept	Cohort	N	α	Target Δ	Random p95	sham
LLaVA-1.5-7B	Effusion	Original	200	-1.00	0.1470	0.1202	0.1546
LLaVA-1.5-7B	Edema	Original	200	-1.00	0.0659	0.1470	0.0913
Qwen2.5-VL-7B	Effusion	Original	200	+0.25	0.2343	0.0963	0.0852
Qwen2.5-VL-7B	Effusion	Independent	400	+0.25	0.1945	0.0878	0.0157

Clinical-direction outcomes				
Model	Concept	Cohort	Clinical margin [95% CI]	Stopped at
LLaVA-1.5-7B	Effusion	Original	—	Random/sham
LLaVA-1.5-7B	Edema	Original	—	Random/sham
Qwen2.5-VL-7B	Effusion	Original	—	Clinical directions
Qwen2.5-VL-7B	Effusion	Independent	-0.0573 [-0.0694, -0.0450]	Clinical directions

establishes selectivity against generic perturbation controls. The same-dose Nodule effect of 0.2863 is descriptive at this stage and motivates the separate prospective specificity endpoint.

4.3 INDEPENDENT ROWS SEPARATE GENERIC SELECTIVITY FROM CLINICAL SPECIFICITY

The independent cohort reproduces Qwen’s generic response (Figure 3). Effusion changes mean $P(\text{yes})$ by 0.1945, above random p95 0.0878 and absolute sham 0.0157. This result uses 400 distinct patients, all absent from the original intervention cohort, and retains the discovery-locked dose.

The clinical comparison reverses the interpretation of that generic pass. Nodule changes mean $P(\text{yes})$ by 0.2519 and realizes the maximum unrelated-direction effect. The reported Effusion-minus-maximum-unrelated margin is -0.0573 . When the maximum over all five fixed clinical directions is recomputed in each bootstrap replicate, its two-sided 95% CI is $[-0.0694, -0.0450]$ and its one-sided 95% lower bound is -0.0678 . The direction-specific conjunction therefore fails despite replicated random/sham selectivity.

4.4 PROSPECTIVE MECHANISM GATES CONFIRM THE ATTRIBUTION BOUNDARY

The causal-ownership matrix provides no named replacement for the failed Effusion attribution. All six diagonal ownership margins are negative, so zero of six clinical concepts is owned under the

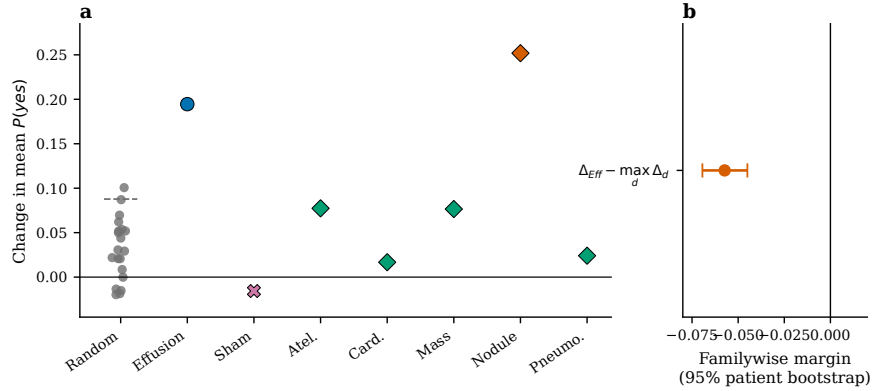


Figure 3: **Qwen Effusion replicates above generic controls but not fixed clinical directions.** (a) Changes in mean $P(\text{yes})$ at $\alpha = +0.25$ on 400 patients absent from the original cohort. The random group contains 20 directions; its dashed segment marks p95. Diamonds show the five fixed unrelated clinical normals. (b) The Effusion-minus-maximum-unrelated statistic with its two-sided 95% patient-bootstrap interval; the registered one-sided 95% lower bound is -0.0678 . Nodule realizes the point-estimate maximum.

Table 3: Prospective Qwen mechanism gates at `vis.last`. Ownership crosses six probe normals with six matching questions; input closure writes each patient’s measured Consolidation displacement back into the negative study. LCB denotes a one-sided simultaneous or familywise 95% lower confidence bound.

Gate	Cohort	Target statistic	Registered comparator	Uncertainty boundary	Outcome
Ownership	400 patients	Alias candidate 0.2909	Random max 0.3402	Alias LCB 0.2808	0/6 owned; no alias
Input closure	50 pairs	$G_C = 0.0022$	Random max 0.0025	γ LCB -0.0364; M LCB -0.0015	No closure

simultaneous and random-control rules. Every question column’s strongest clinical direction is off-diagonal. Effusion is the strongest shared-alias candidate: its mean off-diagonal effect is 0.2909 with simultaneous lower bound 0.2808, but the global maximum across all six questions and 119 random directions is larger at 0.3402. The matrix therefore confirms neither clinical ownership nor a shared alias.

The Consolidation fallback separates availability from input closure on 50 fresh within-patient pairs. On the full patient-disjoint test split, its probe selectivity is 0.0642 with 95% CI $[0.0302, 0.0976]$, so the readout is eligible before pair selection. Across the 50 pairs, mean positive-minus-negative displacement along the Consolidation normal is 0.1091, but its one-sided lower bound is -0.0364 . Writing the measured displacement back produces mean closure gain 0.0022, below the maximum over 119 random directions, 0.0025. The familywise Consolidation-minus-maximum-clinical lower bound is also negative at -0.0015 . The input-closure conjunction is not met.

4.5 THE EVIDENCE CHAIN

The results form one ordered finding. Every target is available to the same-capacity linear readout. The two LLaVA probe normals stop at random/sham, while Qwen Effusion passes generic controls twice but stops at fixed clinical directions. The prospective matrix does not recover a named owner or alias, and the paired fallback does not recover input closure. Together these tests establish that availability and generic perturbation sensitivity do not suffice for concept-specific causal attribution under these final-block interventions.

5 DISCUSSION AND LIMITATIONS

A hierarchy of evidence. The experiments support treating linear availability, generic intervention sensitivity, clinical direction specificity, and mechanism confirmation as different evidentiary

levels. A positive probe interval answers whether a restricted readout reliably recovers a label under the patient split and control task. A random/sham pass answers whether the target normal produces an effect beyond matched non-semantic vectors. Clinical-direction comparisons test whether that response belongs preferentially to the named finding; ownership and input closure then test the proposed read-write account. Collapsing these questions turns the original Qwen response into an attribution, whereas the full sequence supplies no registered clinical owner, alias, or closure effect.

Implication for causal interpretation. Matched clinical directions are a load-bearing control when a probe normal carries a semantic name. Random vectors test the ambient geometry, and a coordinate permutation tests one construction artifact, but neither represents a plausible competing clinical direction learned with the same readout and data. For the Qwen cell, random/sham controls isolate a non-generic writer but do not identify an Effusion-specific one. The practical rule is therefore to compare semantic directions at the same locus, dose, rows, and endpoint, then test the implied mechanism with multiplicity handled over each fixed family.

Scope of the evidence. The registered evidence covers NIH ChestX-ray14, two 7B architectures, the consumed final visual block, Effusion in both models, Edema in LLaVA, six Qwen clinical questions, a paired Consolidation fallback, fixed yes/no prompts, linear readouts, and relative-token probe-normal interventions. It establishes an availability-versus-attribution dissociation for these cells and protocols. The paired decoding intervals span zero, so they leave architecture and concept ordering unresolved. Intermediate loci, nonlinear directions, and other medical image modalities define separate tests rather than extensions of the present estimate.

Data and endpoint limitations. ChestX-ray14 labels are mined from reports, with 179 positive Edema test images versus 700 positive Effusion images. The patient split prevents identity leakage but does not remove label noise or acquisition correlations. Mean pooling also does not localize a finding spatially. The behavioural endpoint is the probability assigned to a yes answer under fixed prompts, so the evidence concerns this answer pathway rather than diagnostic accuracy or report quality.

Specificity and closure design. The independent supplement tests one discovery-locked Qwen dose against five prospectively fixed clinical normals, and the ownership matrix expands the same comparison to six matching questions. Their maxima are recomputed inside the registered bootstrap families. A broader clinical vocabulary could contain stronger or more closely related directions. In the closure gate, the measured displacement produces $|\alpha|$ of order 10^{-4} with no capped pairs; its displacement lower bound crosses zero, and the resulting near-zero gains are not separable from random or clinical controls. The nonconfirmation therefore bounds this registered read-write construction rather than all causal mechanisms in the model.

Reproducibility and ethics. Appendix A records prompts, row-selection hashes, seeds, doses, model revisions, run identifiers, and GPU wall time. Immutable snapshots and terminal receipts bind each result to its environment and inputs. This work evaluates interpretability evidence, not clinical diagnostic performance. Its safety implication is methodological: a decodable feature should be treated as clinically operative only after intervention controls establish direction specificity.

6 CONCLUSION

At the consumed final visual block, all three registered cells make their target label available to a fixed linear readout. The two LLaVA cells stop at the random/sham intervention rung. Qwen Effusion clears that rung twice, but a fixed Nodule direction remains reliably stronger; a subsequent six-question matrix yields no owned concept or shared alias beyond random directions. The paired Consolidation fallback is likewise decodable but does not close the input gap beyond its controls. Under these registered protocols, linear decodability does not establish concept-specific causal use. Clinical interpretation of a decoded direction requires matched specificity and mechanism tests at the same representation, dose, rows, and behavioural endpoint.

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A REGISTERED PROTOCOL AND PROVENANCE

Prompts. The Effusion prompt is “Is there a pleural effusion in this chest radiograph? Answer yes or no.” The Edema prompt is “Is there pulmonary edema in this chest radiograph? Answer yes or no.” The Consolidation prompt substitutes “consolidation” in the same template. The ownership matrix likewise uses one fixed yes/no prompt for each of its six named findings. LLaVA uses `llava-hf/llava-1.5-7b-hf` at revision `b234b804b114d9e37bb655e11cbbb5f5e971b7a9`. Qwen uses `Qwen/Qwen2.5-VL-7B-Instruct` at revision `cc594898137f460bfe9f0759e9844b3ce807cfb5`, with processor resolution fixed at 336^2 pixels.

Rows and seeds. The deterministic NIH split hashes patient identity. Probe projection and fitting use seed 0. Type-to-label controls use seeds 0–19. Probe intervals use 2,000 patient-cluster draws with seed 20260827. The independent Qwen cohort excludes every patient in the original intervention set, chooses one row per remaining patient with SHA-256 of `qwen-direction-specificity-row-v1\0<row_id>`, ranks patients with SHA-256 of `qwen-direction-specificity-patient-v1\0<patient_id>`, and takes the first 400. Its canonical row-list SHA-256 is `76be1da9659973d44a0898fa69c579afc4649f329ee227b19d1216efaa1c8bda`; supplement bootstraps use seed 20260903. The ownership matrix selects a third label-blind 400-patient cohort with row-list SHA-256 `e657a9bd1707f13695b02a341047af6f53981295b840376eb0ea5bbef2d312` and bootstrap seed 20260904. The Consolidation cohort excludes all 1,000 earlier Qwen rows and hash-selects one positive/negative pair for 50 patients; its pair-list SHA-256 is `20dcc8cc3aaa8c65a6f7f759a0ff6bf3b0cf4f1b87f4b65cdc6a15a46c6d85e2`, with bootstrap seed 20260905.

Immutable runs. The accepted full runs are `20260902T191411Z-ffd523c464c8-f99e2f39` (LLaVA Effusion), `20260902T221534Z-110b84618d1b-edema` (LLaVA Edema), `20260903T000321Z-caaa3ef346d-qwen-full` (Qwen Effusion), `20260903T103508Z-456c81bad460-7a3edc4b` (Qwen direction specificity), `20260904T125710Z-a3bd883540eb-causal-ownership` (Qwen ownership), and `20260904T162317Z-8a55a4c2f6b6-consolidation-closure` (Qwen input closure). Their recorded GPU wall times are 7,022, 1,823, 2,077, 361, 9,211, and 445 seconds, respectively, for 20,939 seconds total on one A100 per run. Downstream gates reuse accepted activations or directions where registered.

B COMPLETE REGISTERED SETTINGS

Table 4: Fixed settings shared by the registered cells.

Component	Registered value
Readout	512-dimensional Gaussian projection; logistic regression, $C = 1$
Probe seed	0
Control labels	20 recurring-type maps, seeds 0–19
Probe uncertainty	2,000 patient-cluster bootstrap draws
Concept doses	$-1, -0.5, -0.25, -0.1, 0, 0.1, 0.25, 0.5, 1$
Controlled doses	$-1, -0.5, -0.25, 0.25, 0.5, 1$
Generic controls	20 isotropic random directions; coordinate-permutation sham
Clinical controls	Atelectasis, Pneumothorax, Cardiomegaly, Mass, Nodule
Supplement	$\alpha = +0.25$; 400 patients; 5,000 patient bootstraps
Ownership	6×6 clinical grid; 119 random directions; 5,000 bootstraps
Input closure	50 patient pairs; displacement-matched dose; 5,000 bootstraps

Hook witnesses. For LLaVA, the accepted hook captures `encoder.layers.22` once, preserves a bitwise no-op at $\alpha = 0$, and yields maximum absolute changes of 1.1875 at the downstream connector and 0.1875 in the final logits under a nonzero relative-token intervention. For Qwen, the accepted

hook verifies 32 visual blocks, captures `model.visual.blocks.31`, preserves the $\alpha = 0$ no-op, and changes both `model.visual.merger` and final logits under $\alpha = +0.1$.

C PAIRED DECODING CONTRASTS

Table 5: Paired controlled-decoding contrasts. Both intervals span zero.

Contrast	Estimate	95% patient-bootstrap CI
LLaVA Edema – LLaVA Effusion	0.0221	[−0.0175, 0.0590]
Qwen Effusion – LLaVA Effusion	0.0026	[−0.0155, 0.0215]